# A native spherical CMC/Dirac module for AthenaK

Equations, implementation, and verification

Research implementation for the HengruiZhu99/AthenaK fork

25 October 2026

**Abstract**

The `spherical_cmc` physics module implements a vacuum, spherically symmetric constrained evolution on compactified constant-mean-curvature slices. The Hamiltonian and momentum constraints and CMC lapse are solved through their radial first integrals; a differential elliptic shift controls the metric-derived Dirac connection. Exact boundary values keep future null infinity fixed and null. The module uses AthenaK's mesh, Runge–Kutta coefficients, task graph, outputs, and restart infrastructure, and reuses the fork's hyperboloidal tensor kernels. Its present scope is one radial block, double precision, and host computation. It is a numerical research prototype, not a freely evolving Z4c system or a proof of the earlier proposed three-dimensional hyperboloidal initial-boundary problem.

## 1 Scope and relation to the existing fork

The implementation branches from `z4c_hyperboloidal`, revision `ca77b353a60a` (25 October checkout). It adds an exclusive physics module, selected by an input block `<spherical_cmc>`, without changing the existing Cauchy Z4c or hyperboloidal evolution choices. The new native module is distinct from the pre-existing standalone spherical driver.

Spherical vacuum gravity has no radiative degrees of freedom. Here the mass and CMC momentum integration constant are fixed, while a nonstationary radial coordinate distortion evolves under an elliptic gauge. The two supported families are regular-center Minkowski geometry and excised Schwarzschild CMC trumpets. This is a useful test of scri limits, constrained stage projection, elliptic gauge evolution, and black-hole excision; it is not a gravitational-wave or matter-collapse test. A lapse pulse independent of the constraints is not an allowed datum in this formulation.

The design follows the CMC/conformal-Dirac direction of the earlier report. Published constrained hyperboloidal formulations provide motivation [2, 3], while CMC/spatial-harmonic well-posedness on other domains [4] cannot be transferred without a new scri boundary argument.

## 2 Notation and the change from ordinary Z4c

Following the physical and spatial-conformal notation of Daszuta et al., Sec. 3 [1], use signature $(-,+,+,+)$, $K_{ij} = -\frac{1}{2}\mathcal{L}_n\gamma_{ij}$, and

$$\chi = \psi^{-4}, \qquad \tilde{\gamma}_{ij} = \chi\gamma_{ij}, \qquad \tilde{A}_{ij} = \chi(K_{ij} - \tfrac{1}{3}K\gamma_{ij}), \qquad \widehat{K} = K - 2\Theta. \tag{1}$$

Bars denote a separate Penrose compactification, not the spatial conformal transformation:

$$\bar{g}_{ab} = \Omega^2 g_{ab}, \quad \bar{\alpha} = \Omega\alpha, \quad \bar{\chi} = \Omega^{-2}\chi, \quad \bar{\tilde{A}}_{ij} = \Omega^{-1}\tilde{A}_{ij}. \tag{2}$$

The module fixes $\Omega = (1-r^2)/2$, $0 \le r \le 1$, $K = -3$, and units in which the scri radius and CMC curvature scale equal one. The shift is unchanged by Penrose rescaling. In Cartesian tensor components, with $n_i = x_i/r$,

$$\tilde{\gamma}_{ij} = b\delta_{ij} + (a-b)n_i n_j, \qquad b = a^{-1/2}, \qquad \det\tilde{\gamma} = ab^2 = 1. \tag{3}$$

Thus the spherical line element is $\bar{\gamma}_{ij}dx^i dx^j = \bar{\chi}^{-1}(a\,dr^2 + br^2 d\omega_2^2)$. The code variable `scmc_a` is $a$; it is not a curvature length scale.

The changes from free Z4c evolution are explicit:

| Ordinary Z4c quantity | Treatment in this module |
|---|---|
| $\tilde{\gamma}_{ij}$ | Only its spherical radial eigenvalue $a$ is evolved; determinant one is algebraic. |
| $\chi, \tilde{A}_{ij}$ | Replace their evolution by radial vacuum constraint reconstruction of $\bar{\chi}, \bar{\tilde{A}}_{ij}$. |
| $\widehat{K}, \Theta$ | Set $\widehat{K} = -3$, $\Theta = 0$ at every stage. |
| Evolved connection | Replace by the metric-derived contracted conformal connection; $Z_i = 0$ is imposed. |
| Lapse driver | Use the CMC lapse solution, with $\bar{\alpha}\vert_{\mathscr{I}^+} = 1$. |
| Gamma driver | Replace by the elliptic Dirac-relaxation shift below. |
| Tangential shift | Zero by spherical symmetry. |

The projected variables are stored for native output and restart, but are not independent evolved degrees of freedom. Existing full tensor equations are used as independent differential compatibility diagnostics. This distinction is essential when interpreting constraint propagation and hyperbolicity.

## 3 Solving the constraints and the CMC lapse

Let $R$ be the physical areal radius, $U = 1/R$, and

$$D(R) = 1 - \frac{2M}{R} + \left(-R + \frac{C}{R^2}\right)^2, \qquad F(U) = U^2 D(1/U) = U^2 - 2MU^3 + (-1 + CU^3)^2. \tag{4}$$

In areal coordinates the physical spatial metric is $dR^2/D + R^2 d\omega_2^2$. The trace-free physical extrinsic curvature has mixed eigenvalues $(-2C/R^3, C/R^3, C/R^3)$. Hence

$$^{(3)}R = \frac{2(1-D-RD')}{R^2} = -6 + \frac{6C^2}{R^6}, \tag{5}$$

$$A_{ij}A^{ij} = \frac{6C^2}{R^6}, \qquad {}^{(3)}R - A_{ij}A^{ij} + \tfrac{2}{3}K^2 = 0, \tag{6}$$

$$\partial_R(-2C/R^3) + \frac{2}{R}(-2C/R^3 - C/R^3) = 0. \tag{7}$$

The last equation is the radial momentum constraint.

Matching the angular and radial metric components gives

$$\frac{dR}{dr} = \frac{a^{3/4}R\sqrt{D}}{r}, \qquad \tau(r) = \int_r^1 \frac{a(s)^{3/4}}{s}\,ds = -\log r + J(r), \qquad J(r) = \int_r^1 \frac{a(s)^{3/4}-1}{s}\,ds. \tag{8}$$

Consequently $dU/d\tau = \sqrt{F(U)}$, $U(\tau = 0) = 0$ is a regular scri initial value. The reconstructed fields, for $r < 1$, are

$$\bar{\chi} = b\left(\frac{rU}{\Omega}\right)^2, \qquad \bar{\tilde{A}}_{rr} = -\frac{2aCU^3}{\Omega}, \tag{9}$$

$$c_t = a(1)^{3/4}, \qquad \bar{\alpha} = c_t\Omega\frac{\sqrt{F}}{U}, \qquad \beta^r_{\mathrm{stat}} = c_t r a^{-3/4}(-1 + CU^3). \tag{10}$$

The transverse eigenvalue of $\bar{\bar{A}}_{ij}$ is $-b\bar{\bar{A}}_{rr}/(2a)$, so its conformal trace vanishes identically. The radial component is a Cartesian radial eigenvalue, as used by the existing spherical tensor kernel.

The physical CMC lapse equation is

$$-\Delta_\gamma \alpha + (3 + 6C^2/R^6)\alpha = 0. \tag{11}$$

Its selected solution is $\alpha = c_t\sqrt{D}$. Indeed $\Delta_\gamma\sqrt{D}/\sqrt{D} = D''/2 + D'/R = 3 + 6C^2/R^6$. An excision boundary would allow other homogeneous lapse choices; this module selects the lapse inherited from the fixed trumpet, rather than claiming a uniqueness theorem from the scri value alone. No lapse matrix solve is necessary for this restricted vacuum family.

### 3.1 Regular-center family

Smoothness requires $a(0) = 1$ and $a - 1 = O(r^2)$. Regularity selects $M = C = 0$ and $U = \sinh\tau$. The expression evaluated in the code is

$$\bar{\chi} = b\left[e^{-J}\left(1 + \frac{\mathrm{expm1}(2J)}{1 - r^2}\right)\right]^2, \quad \bar{\alpha} = c_t\sqrt{\Omega^2 + r^2 b/\bar{\chi}}, \quad \beta^r_{\mathrm{stat}} = -r\left[\frac{a(1)}{a(r)}\right]^{3/4}. \tag{12}$$

This avoids forming $U = \infty$ at the origin. The reference $a = 1$ has $\bar{\chi} = 1$, $\bar{\alpha} = (1 + r^2)/2$, $\beta^r = -r$, and vanishing trace-free curvature. The lapse calculation uses a rationalized difference from this reference to preserve the exact stationary state to roundoff.

### 3.2 Excised black-hole family

For $M > 0$ use $r_{\mathrm{in}} > 0$ and the critical $C$ supplied by the fork's existing `CMCTrumpet`. It solves for a double zero of $D$ at the trumpet radius and evaluates $\sqrt{F}$ in factored form, avoiding cancellation near that zero. The new public method `InverseArealDerivative` exposes that existing first integral for reuse. Radial integration uses RK4 with conformal-distance substeps at most 0.002. The inner boundary must satisfy $R_{\mathrm{in}} < 2M$ and have purely inward physical characteristic flow. The example uses $M = 0.5$, $r_{\mathrm{in}} = 0.05$. Neither a puncture floor nor evaluation at $r = 0$ is used for black holes.

## 4 Elliptic Dirac gauge and metric evolution

Write $u = 1/a$ and let $V$ be the radial coefficient of the metric-derived contracted conformal connection:

$$\widehat{\Gamma}^i = Vn^i, \qquad V = -u' - \frac{2(u - u^{-1/2})}{r}. \tag{13}$$

This connection is not a separately evolved Z4 variable. The gauge prescribes

$$\partial_t V = -\eta V, \quad \eta \geq 0, \qquad \beta^r = \beta^r_{\mathrm{stat}} + ry. \tag{14}$$

The static shift and reconstructed extrinsic curvature cancel exactly in the continuum metric equation. Its remaining part is

$$\partial_t a = r\left(ya' + \tfrac{4}{3}ay'\right), \qquad W := \partial_t u = ryu' - \tfrac{4}{3}ury'. \tag{15}$$

Taking the variation of (13) gives $V_t = -W' - cW$, where $c = (2 + u^{-3/2})/r$. Direct substitution yields the scalar elliptic equation

$$y'' + Py' + Qy = -\frac{3\eta V}{4ur}, \tag{16}$$

$$P = \frac{u'}{4u} + \frac{1}{r} + c, \qquad Q = -\frac{3}{4u}\left[u'' + u'\left(\frac{1}{r} + c\right)\right]. \tag{17}$$

The coefficient of $y''$ is one, independent of the metric and of $\Omega$. Boundary conditions are

$$y(1) = 0, \qquad \begin{cases} y'(0) = 0, & M = 0, \\ y(r_{\rm in}) = 0, & M > 0. \end{cases} \tag{18}$$

At a regular center $P = 4/r + O(r)$ and $Q$ and $V/r$ are finite. The leading operator is the radial Laplacian in five dimensions. Even reflection supplies the regular branch without evaluating $1/r$ at $r = 0$.

Using $y$ is important numerically: evaluating $v' - v/r$ for $v = ry$ using unrelated finite differences leaves a truncation error at the first cell that need not respect $a_t = O(r^2)$. The form (15) preserves the required radial scaling. This corrected a nonconvergent center error found during development.

There is a useful continuum uniqueness argument beyond the flat reference. For the homogeneous shift problem $V_t = 0$, factorization gives

$$W' + cW = 0, \qquad (u^{-3/4}y)' = -\frac{3}{4r}u^{-7/4}W. \tag{19}$$

On the ball, $c = 3/r + O(r)$, so a nonzero homogeneous $W$ behaves as $r^{-3}$. Regularity excludes it. Then $y$ is proportional to $u^{3/4}$, and $y(1) = 0$ sets that constant to zero. On an excised annulus write $W = A\exp(-\int c\,dr)$. Integrating the second equation between two zero Dirichlet values gives $A$ times a strictly positive integral equal to zero. Thus $A = 0$ and again $y = 0$. This proves absence of a homogeneous kernel for smooth positive $u$ satisfying the stated center regularity or annular boundary conditions. Standard elliptic Fredholm theory then gives invertibility: on the ball the operator is understood as a smooth radial five-dimensional elliptic operator, since $P - 4/r = O(r)$ and $Q$ is regular. Bounds require positive lower bounds on $u$, controlled smooth coefficients, and a fixed annular inner radius; no uniform estimate as those assumptions degenerate is asserted.

The same identities explain the reduced continuum dynamics. $V(t, r)$ is known as $e^{-\eta t}V(0, r)$; recovering $u$ solves the first-order equation (13). On the ball its regular branch fixes the integration constant. On an annulus the conserved conformal radial distance $I = \int_{r_{\rm in}}^1 u^{-3/4}dr/r$ fixes it, since $I_t = [u^{-3/4}y]_{r_{\rm in}}^1 = 0$. The derivative of $I$ with respect to the inner value of $u$ is strictly negative: the corresponding first-order variation is $z' = -cz$, with $z(r_{\rm in}) = 1$, so $I' = -\frac{3}{4}\int u^{-7/4}z\,dr/r < 0$. This gives a constructive local smooth solution on a positive branch via the implicit function theorem, rather than a hyperbolic characteristic argument. It is specific to the fixed-mass spherical constraint manifold. The code checks pivots, finite values and the original matrix residual as additional discrete safeguards; continuum invertibility alone does not validate a discretization.

Linearizing on the regular-center reference, $a = 1 + \epsilon f$ gives $V = \epsilon(f' + 3f/r) + O(\epsilon^2)$. Thus $(\partial_r + 3/r)(f_t + \eta f) = 0$ and regularity implies $f_t = -\eta f$. This proves bounded damping of this reduced linear gauge degree of freedom. It does not analyze the full off-constraint Z4c symbol. On an excised annulus there is an additional homogeneous radial distortion, and the two shift boundary conditions determine its evolution; simple pointwise damping of $a - 1$ is not expected there.

## 5 Scri fixing, regularity, and boundary interpretation

At $r = 1$ the analytic limits are imposed as exact assignments:

$$\bar{\chi} = a(1), \qquad \bar{\alpha} = 1, \qquad \beta^r = -1, \qquad \bar{\tilde{A}}_{rr} = 0. \tag{20}$$

Since $U = a(1)^{3/4}(1 - r) + O((1 - r)^2)$ and $\Omega = (1 - r) + O((1 - r)^2)$, these are the limits of (9)–(10), not floors or extrapolated values. With $\partial_t\Omega = 0$,

$$\bar{g}^{ab}\partial_a\Omega\partial_b\Omega = \Omega'^2\left(\frac{\bar{\chi}}{a} - \frac{(\beta^r)^2}{\bar{\alpha}^2}\right) = 0 \quad \text{at } r = 1. \tag{21}$$

Physical radial null coordinate speeds there are $-\beta^r \pm \bar{\alpha}\sqrt{\bar{\chi}/a} = 0, 2$: no physical characteristic enters from beyond scri. At the excision boundary the code checks $-\beta^r + \bar{\alpha}\sqrt{\bar{\chi}/a} < 0$ after every solve.

The gauge is elliptic, so finite-speed hyperbolic gauge characteristics are not its appropriate boundary description. Its boundary data are explicitly fixed by (18) and the lapse choice. Constraints are projected rather than freely propagated; truncation error remains and is measured independently. There are no independent incoming constraint characteristics in this reduced state vector, but this does not prove immunity to numerical constraint errors.

Smooth reconstructed spherical data have finite scri limits. The existing unregularized tensor diagnostic kernel is only evaluated at interior cell centers with $\Omega > 0$; it is never called at the exact boundary. Its cancellations can amplify differentiation error by $1/\Omega$. In particular, a low-order quadrature for $J$, followed by division by $\Omega$ and two derivatives, generated boundary defects during development. The present higher-order quadrature and closures address the tested cases; they do not constitute a general uniform error estimate for arbitrary perturbed data. A naive mass diagnostic close to scri also suffers cancellation, so the history maximum is restricted to $r \leq 0.8$, with the unrestricted field still available for inspection.

## 6 Discretization and native AthenaK integration

There are $N$ ordinary AthenaK radial cell centers and two exact endpoint nodes: $r_i = r_{\rm in} + (i - \frac{1}{2})h$ for $1 \leq i \leq N$, with $h = (1 - r_{\rm in})/N$. The nearest ghost slots store the two endpoint metric degrees of freedom for restart; they are not ordinary field samples at ghost cell coordinates. Other ghost slots are zero. Field output with ghost zones is therefore rejected.

Differential operators have fourth-order spatial accuracy on smooth data: five-point centered formulas in the interior, six-point polynomial closures on the unequal endpoint intervals, and reflected even/odd stencils at a regular center. The PDE remains second order in space. Quintic interpolation and three-point Gaussian quadrature integrate $J$ on each interval. The elliptic matrix has fixed bandwidth; banded elimination uses $O(N)$ work and storage, with no pivoting and explicit small-pivot rejection. The residual is measured against the original, uneliminated operator and must be below $10^{-8}$ after normalization by $1 + |\text{source}|$. A three-dimensional multigrid hierarchy would add unnecessary machinery to this single radial solve.

The native RK3 task sequence is

$$\texttt{CopyState} \rightarrow \texttt{SolveGauge} \rightarrow \texttt{RKUpdate} \rightarrow \texttt{SolveGauge} \rightarrow \texttt{NewTimeStep}.$$

At each stage, projection supplies the fields on which the RHS is evaluated; after the metric update it is repeated. The endpoint $a(1)$ evolves too; fixing it independently would change the elliptic boundary problem. The regular-center endpoint is reset to $a(0) = 1$. The task uses AthenaK's $\gamma_0, \gamma_1, \beta$ RK coefficients and the mesh time step. The reported time-step estimate is

$$\Delta t_{\rm module} = \min\left\{\frac{h}{\max(1, |\beta^r| + \bar{\alpha}\sqrt{\bar{\chi}/a})}, \frac{1}{1+\eta}\right\}, \tag{22}$$

with the native CFL multiplier applied by the mesh. This is an operational conservative choice, not a stability theorem for all nonlinear data.

| File or component | Role |
|---|---|
| `src/spherical_cmc/radial_solver.hpp` | Radial projection, banded elliptic solve, metric RHS, and tensor diagnostics. |

| File or component | Role |
|---|---|
| `src/spherical_cmc/spherical_cmc.hpp/.cpp` | Native arrays, validation, initialization, field packing, time step, and history. |
| `src/spherical_cmc/spherical_cmc_tasks.cpp` | Native RK stage task assembly and updates. |
| `MeshBlockPack` | Owns the module, selects exclusive physics, destroys its state. |
| `Mesh`, `ProblemGenerator` | Native CFL selection and initial/restart setup. |
| `BaseTypeOutput`, history, restart | Field selection, module histories, versioned restart payload. |
| Existing `spherical_tensor.hpp`, `conformal_constraints.hpp`, and conformal RHS | Reused geometry, vacuum constraints, physical mass and full tensor RHS; no analytic reference RHS is subtracted. |
| Existing `cmc_trumpet.hpp` | Critical trumpet constant and factored first integral. |

All radial solves currently execute on the host. Native Kokkos arrays and explicit host/device copies keep the storage interface consistent with other modules. GPU performance, multi-block decomposition, MPI with more than one rank, AMR, zooming, matter coupling, and arbitrary $K$ or scri radius are not supported. Unsupported physics combinations and integrators fail explicitly.

## 7 Inputs, outputs, and reproduction

From the repository root, a serial build and example runs are:

```
cmake -S . -B build-cmc -DCMAKE_BUILD_TYPE=Release \
  -DAthena_ENABLE_MPI=OFF -DKokkos_ENABLE_SERIAL=ON \
  -DAthena_ENABLE_SPHERICAL_CMC_TESTS=ON
cmake --build build-cmc -j4
mkdir -p run-cmc
cd run-cmc
../build-cmc/src/athena -i ../inputs/spherical_cmc/minkowski.athinput
# Or use ../inputs/spherical_cmc/schwarzschild.athinput
```

Both examples use $a(0,r) = \exp[Ar^2(1-r^2)^2]$, with $A = 0.05$, $\eta = 1$. Set `spherical_cmc/metric_pulse=0` for a stationary reference, or `dirac_damping=0` for a stationary distorted coordinate metric. Keep `mesh/nx1` and `meshblock/nx1` equal when changing resolution. The minimum radial resolution is eight cells and at least two ghost slots are required. The mesh ends exactly at $r = 1$.

Native `variable=spherical_cmc` exposes 13 fields:

| Name | Meaning |
|---|---|
| `scmc_a, scmc_chi, scmc_alpha, scmc_beta` | $a, \bar{\chi}, \bar{\alpha}, \beta^r$ |
| `scmc_Khat, scmc_Theta, scmc_Arr` | $-3, 0, \bar{\tilde{A}}_{rr}$ |
| `scmc_Gamma` | Metric-derived $V$ |
| `scmc_H, scmc_Mr` | Independent physical Hamiltonian and radial momentum residuals |
| `scmc_Krhs, scmc_Arhs` | Full tensor RHS for $K$ and $\bar{\tilde{A}}_{rr}$ |
| `scmc_mass_error` | Spherical physical mass diagnostic minus prescribed $M$ |

The $\bar{\tilde{A}}_{rr}$ RHS need not vanish during a black-hole gauge evolution: it must agree with the tangent of the reconstructed $\bar{\tilde{A}}_{rr}$, which a separate test checks. $K_t$ should vanish in the continuum. Exact endpoint diagnostics are in the history file, not in cell-centered tables. Restart retains the endpoint metric values and the ordinary AthenaK time-step state. Dependent fields are recomputed after reading. `state_version=1` identifies the payload layout; changing the physics family or mass during restart is not a supported physical continuation.

## 8 Verification and remaining limits

The reproducible checks are

```
python3 tst/spherical_cmc/check_equations.py
ctest --test-dir build-cmc -R spherical_cmc --output-on-failure
ATHENA_CMC_EXE="$PWD/build-cmc/src/athena" python3 -m pytest -q \
  tst/test_suite/spherical_cmc/test_native_cpu.py
```

The symbolic script verifies the Dirac operator, the physical Hamiltonian and momentum first integrals, the CMC lapse, exact nullness, and the regular-center linear gauge damping identity. The C++ test checks reference preservation, spatial refinement, rejection of nonfinite metrics, and agreement between the constraint-projection tangent and the existing tensor $\bar{\tilde{A}}_{rr}$ equation. The native tests cover finite-amplitude gauge evolution, exact scri nullness, regular-center and black-hole refinement, bitwise restart equivalence (including both native restart file organizations), and invalid configurations.

Representative double-precision runs with $A = 0.05$, $\eta = 1$, $t = 2$ gave:

| Family | $N$ | $\|H\|_\infty$ | $\|K_t\|_\infty$ |
|---|---|---|---|
| Regular center | 32 | $2.03\,10^{-7}$ | $9.61\,10^{-7}$ |
| | 64 | $1.31\,10^{-8}$ | $1.51\,10^{-7}$ |
| | 128 | $9.93\,10^{-10}$ | $2.07\,10^{-8}$ |
| Black hole, $M = 0.5$ | 64 | $1.01\,10^{-3}$ | $7.48\,10^{-6}$ |
| | 128 | $1.85\,10^{-4}$ | $8.67\,10^{-7}$ |
| | 256 | $3.77\,10^{-5}$ | $9.29\,10^{-8}$ |

These are measured errors, not a claim of fourth-order global evolution or uniform convergence in every diagnostic. The black-hole Hamiltonian converges more slowly than the smooth static projection; the regular-center Hamiltonian encounters a floating-point differentiation floor at higher resolution. The null residual at the exact outer boundary is exactly zero in these runs. The native solver uses third-order time integration.

The global source modifications were also checked with the existing Z4c CPU overhaul, binary-conversion and restart regressions. The radial kernel is suitable for address/undefined-behavior sanitizer checks; hardware backend and MPI validation remain future work. Detailed recorded commands and pass counts are in the companion `spherical_cmc.md`.

This module is ready for controlled spherical numerical experiments within the stated scope. It does not establish uniform strong hyperbolicity of an unconstrained three-dimensional Z4c system, nor nonlinear well-posedness of the full conformal elliptic-hyperbolic scri problem. The next scientific extensions are a consistent spherical matter sector (introducing physical dynamics), regular mass extraction at scri, and boundary estimates for the intended three-dimensional formulation. These must precede interpreting spherical success as resolution of the original oblique-direction hyperbolicity problem.

## References


[1] B. Daszuta et al., *GR-Athena++: puncture evolutions on vertex-centered oct-tree AMR*, arXiv:2101.08289, https://arxiv.org/abs/2101.08289.

[2] V. Moncrief and O. Rinne, *Regularity of the Einstein Equations at Future Null Infinity*, arXiv:0811.4109, https://arxiv.org/abs/0811.4109.

[3] O. Rinne and V. Moncrief, *Hyperboloidal Einstein-matter evolution and tails for scalar and Yang–Mills fields*, arXiv:1301.6174, https://arxiv.org/abs/1301.6174.

[4] L. Andersson and V. Moncrief, *Elliptic-hyperbolic systems and the Einstein equations*, arXiv:gr-qc/0110111, https://arxiv.org/abs/gr-qc/0110111.

[5] A. Vañó-Viñuales, S. Husa and D. Hilditch, *Spherical symmetry as a test case for unconstrained hyperboloidal evolution*, arXiv:1412.3827, https://arxiv.org/abs/1412.3827.